

0050 Fundamental Model v0.1 — Stage B Predictive Validation

Research question

Do the already-Frozen Quality, Fundamental State, state_detail and Valuation categories contain information about later adjusted-close returns?

Frozen data contract

Stage A signal SHA-256: 394e584bb83adc6b4a001ef605aa4dd58540fb76b48ddf1013d8d39c5bd2eeb2. Entry is exactly Stage A first_trade_date. Horizons are 60, 120, 252 and 504 common TWSE trading sessions.

Cohort boundary

Only the 38 non-financial issuers in the fixed September 2026 cohort enter the analysis. Twelve financial issuers are excluded. The claim is current fixed cohort predictive association only.

Eligible returns

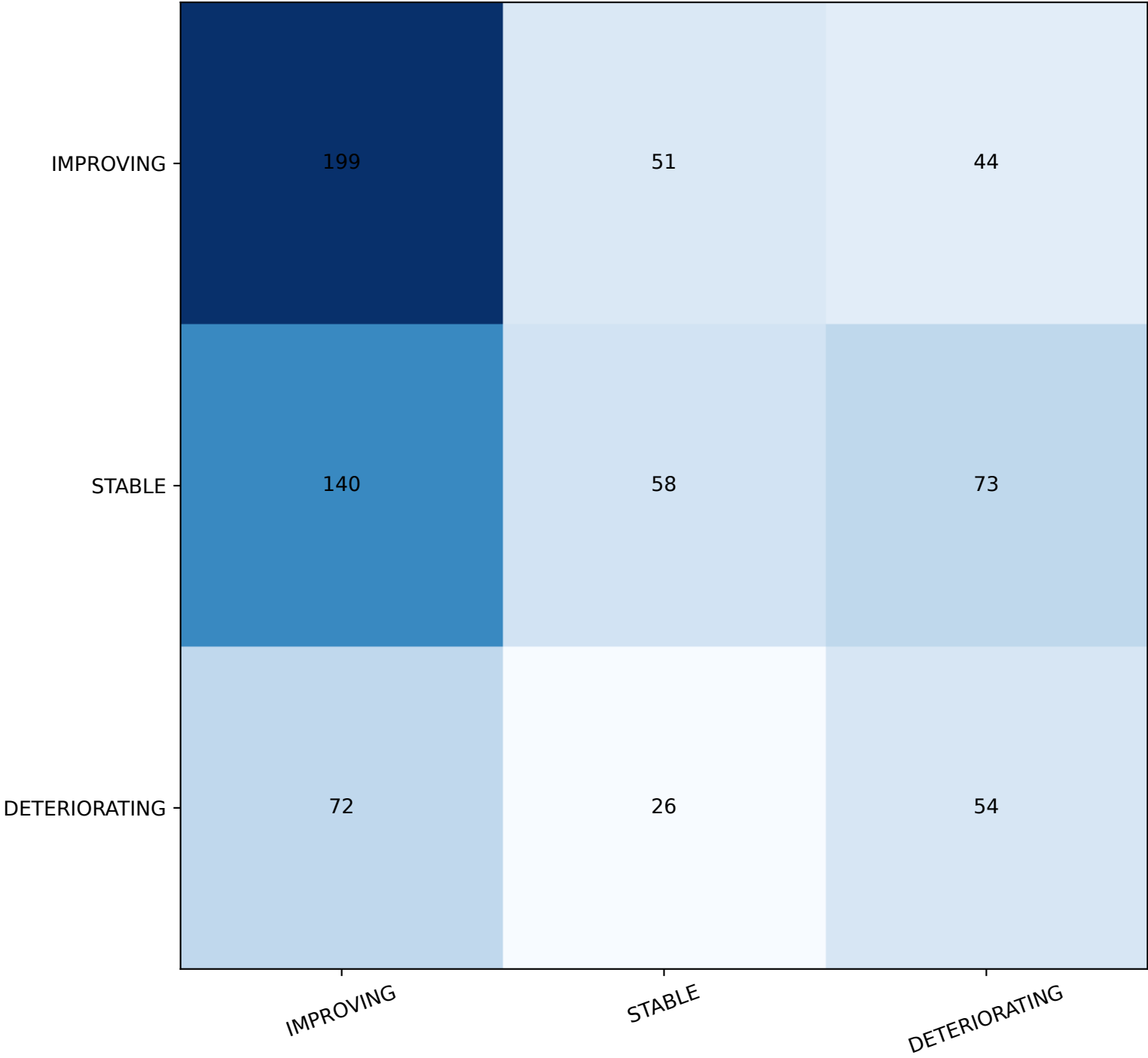
60d: 1507, 120d: 1462, 252d: 1395, 504d: 1224

Conclusion

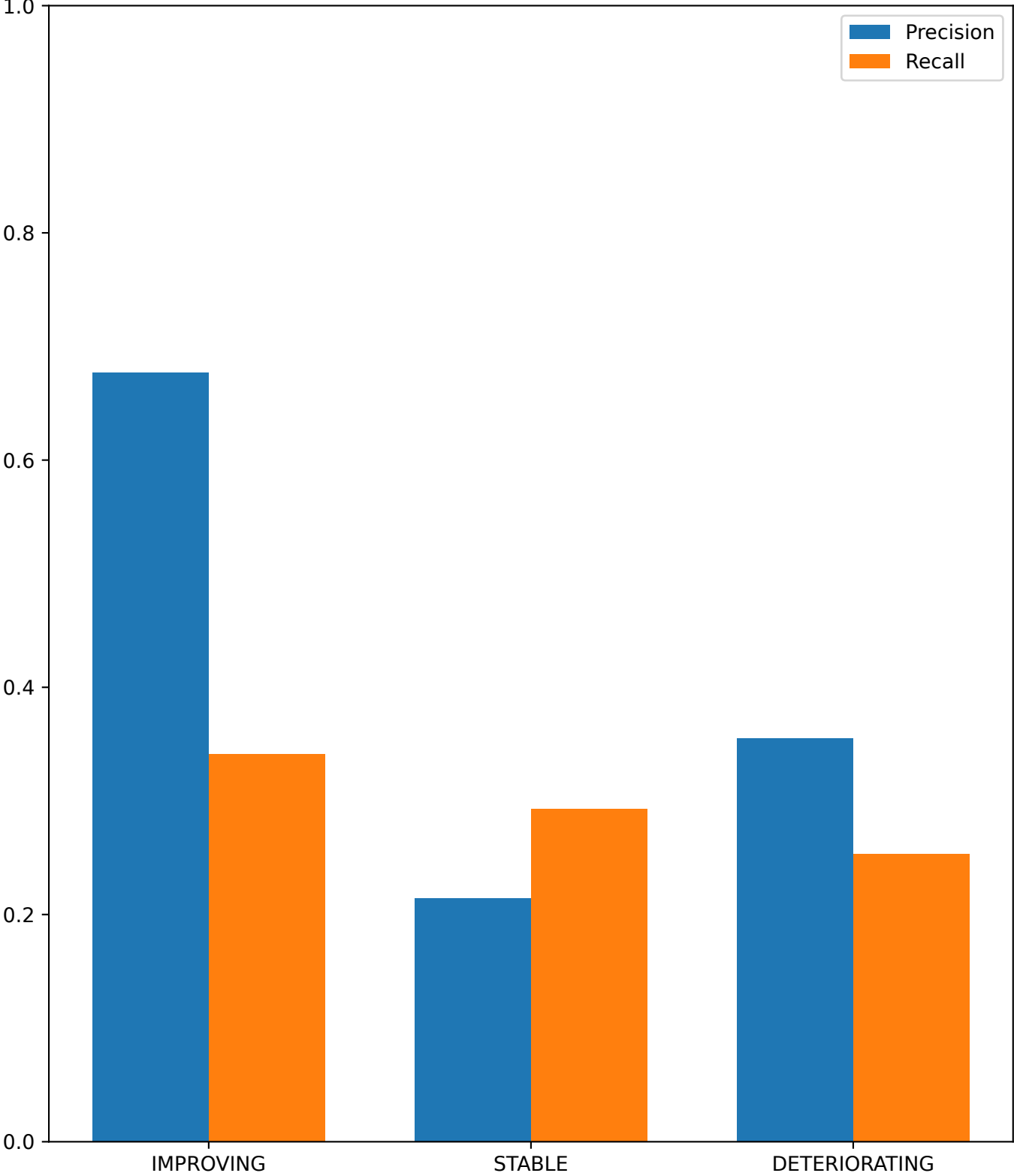
Primary research evidence assessment: MODERATE. Mechanical rubric grade: STRONG. Independent reviewer assessment: MODERATE. Label: EXPLORATORY_PREDICTIVE_ASSOCIATION.

A. Fundamental Identification Accuracy

Frozen identification confusion matrix



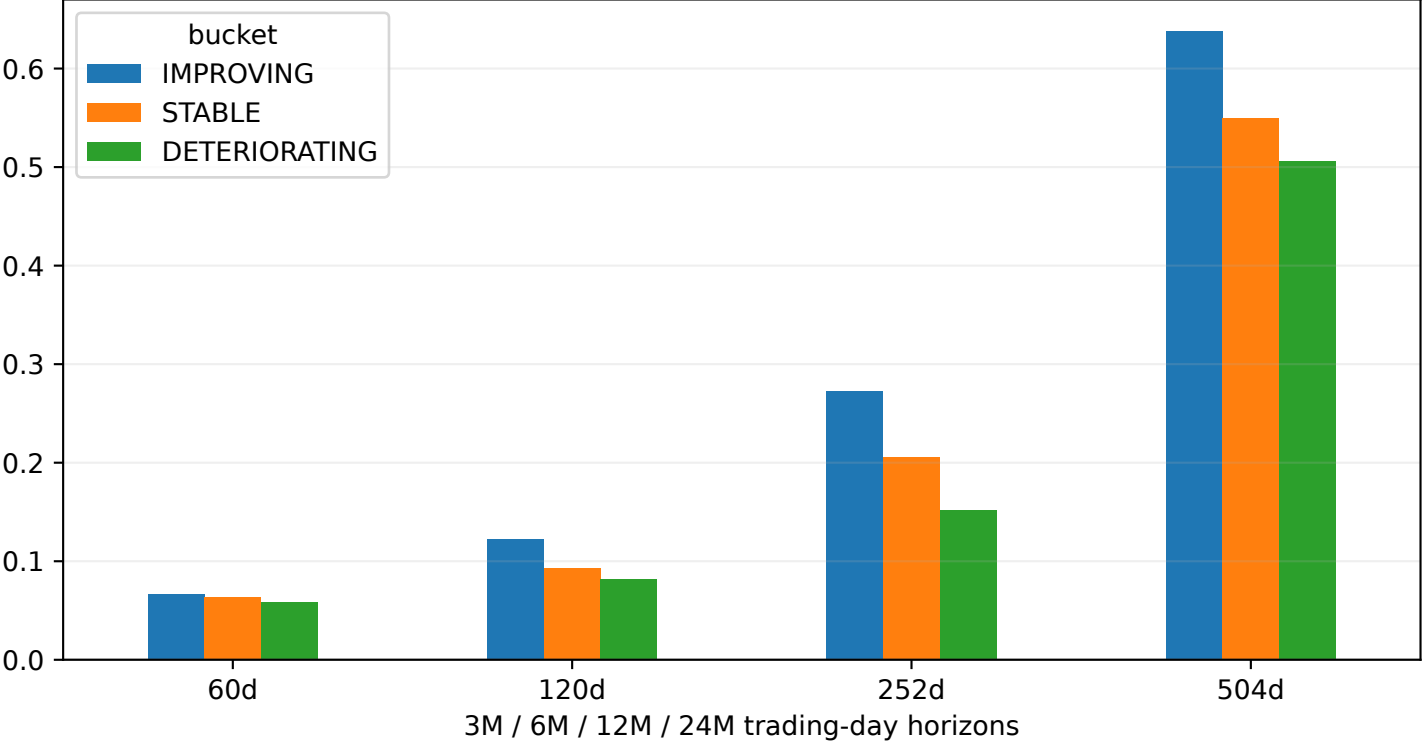
Frozen canonical-state accuracy



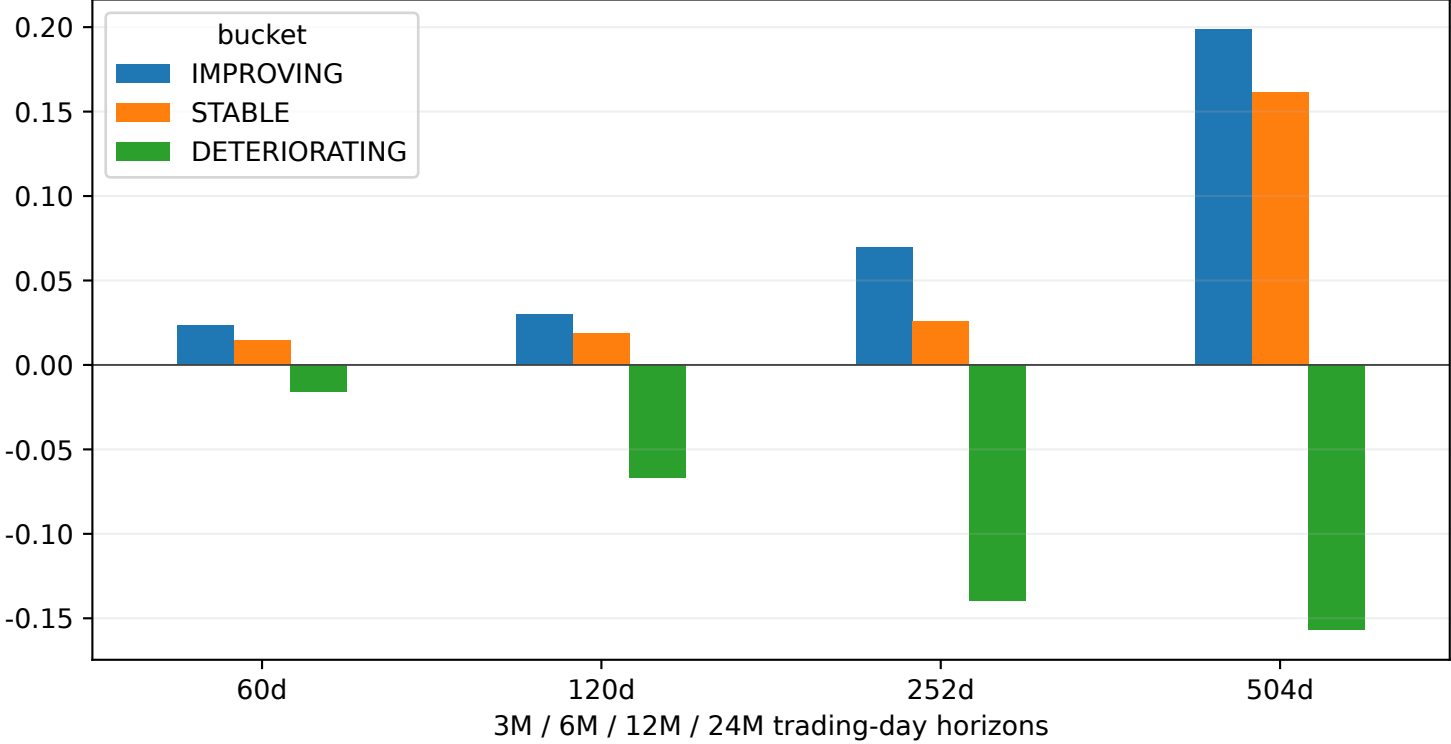
Identification accuracy and future-return predictive power are separate research layers.

B1. Future Return Predictive Power — Canonical State

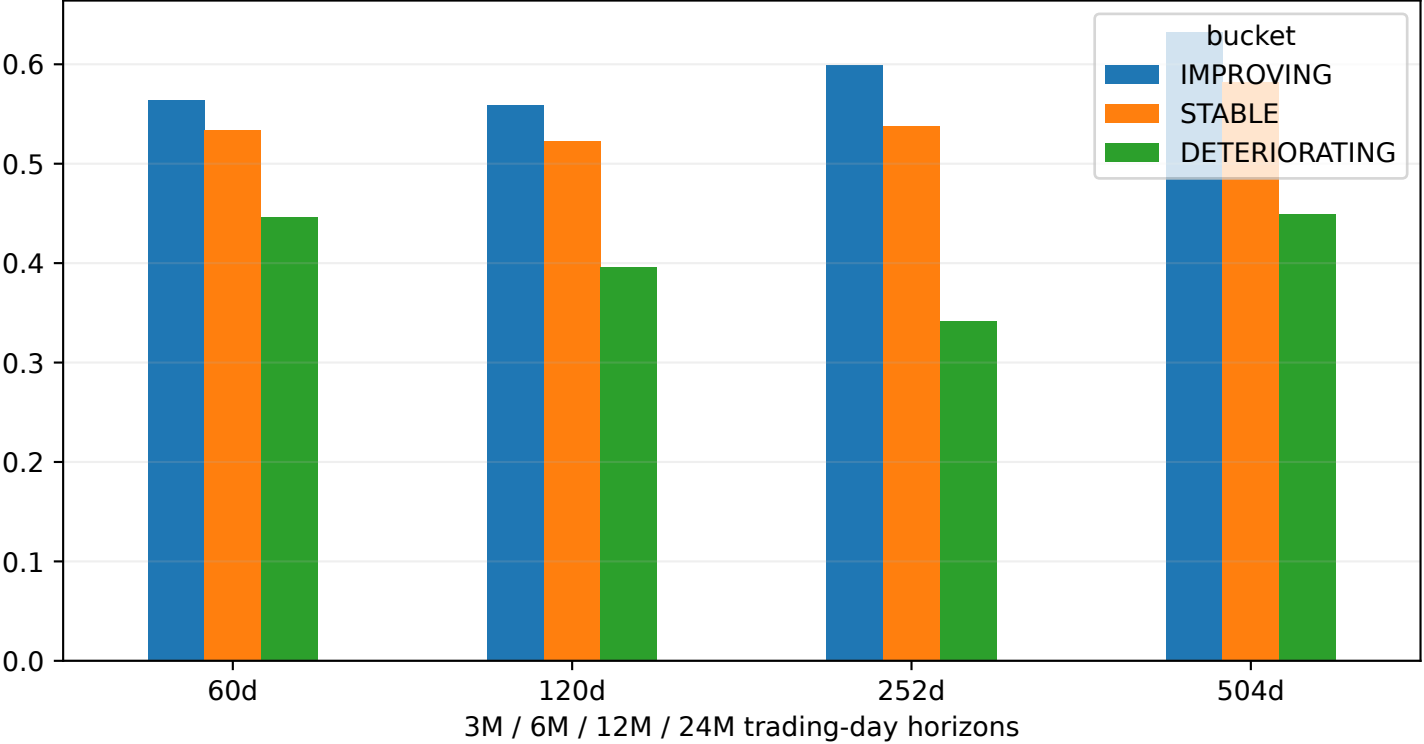
Median stock return



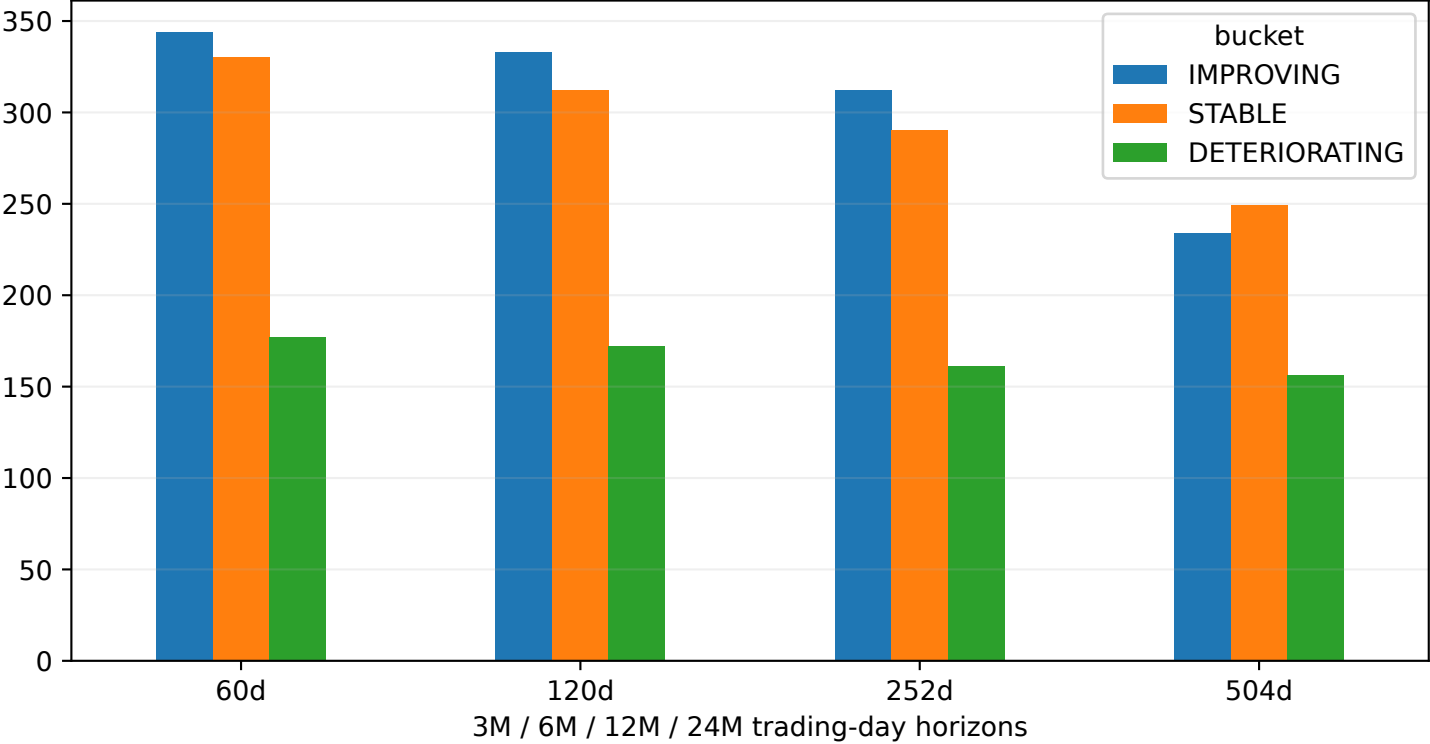
Median excess return vs 0050



Outperform-0050 rate



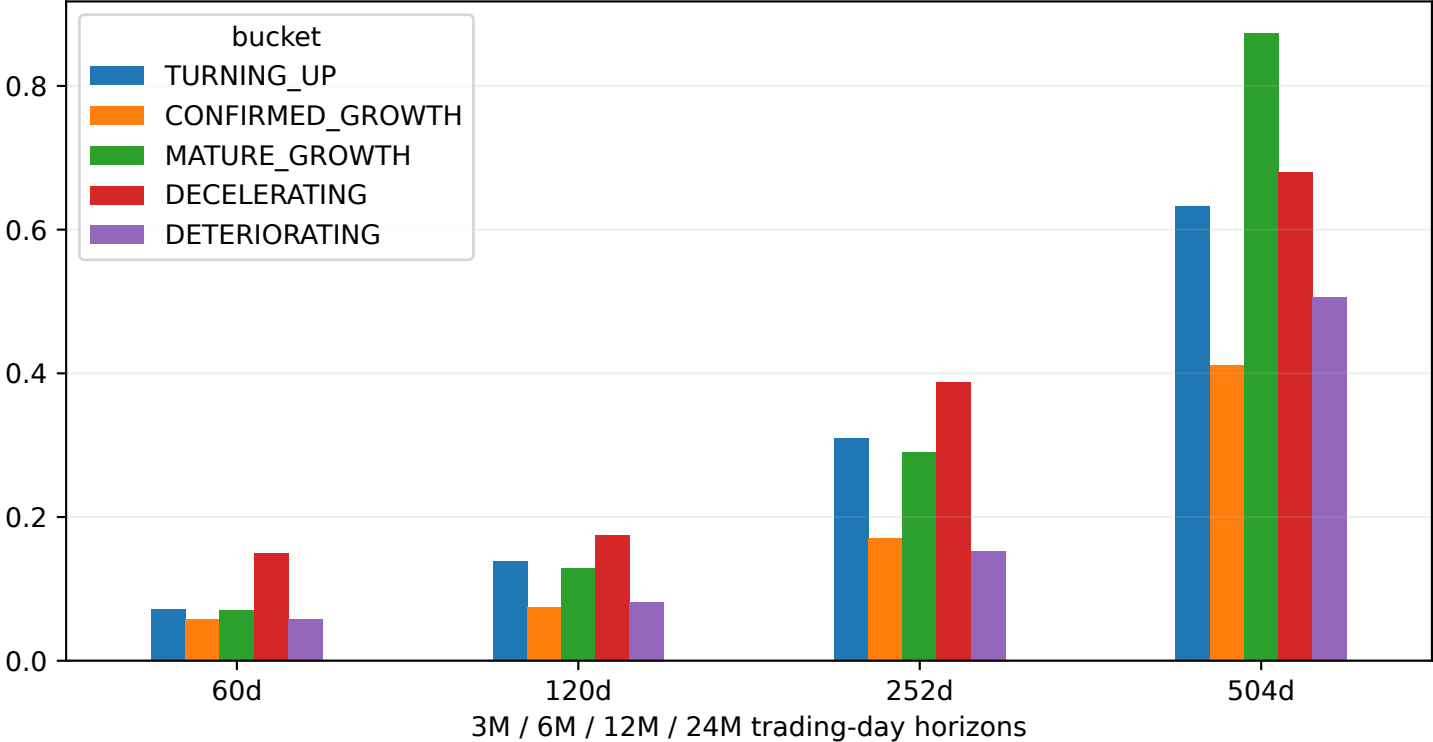
Evaluable observations



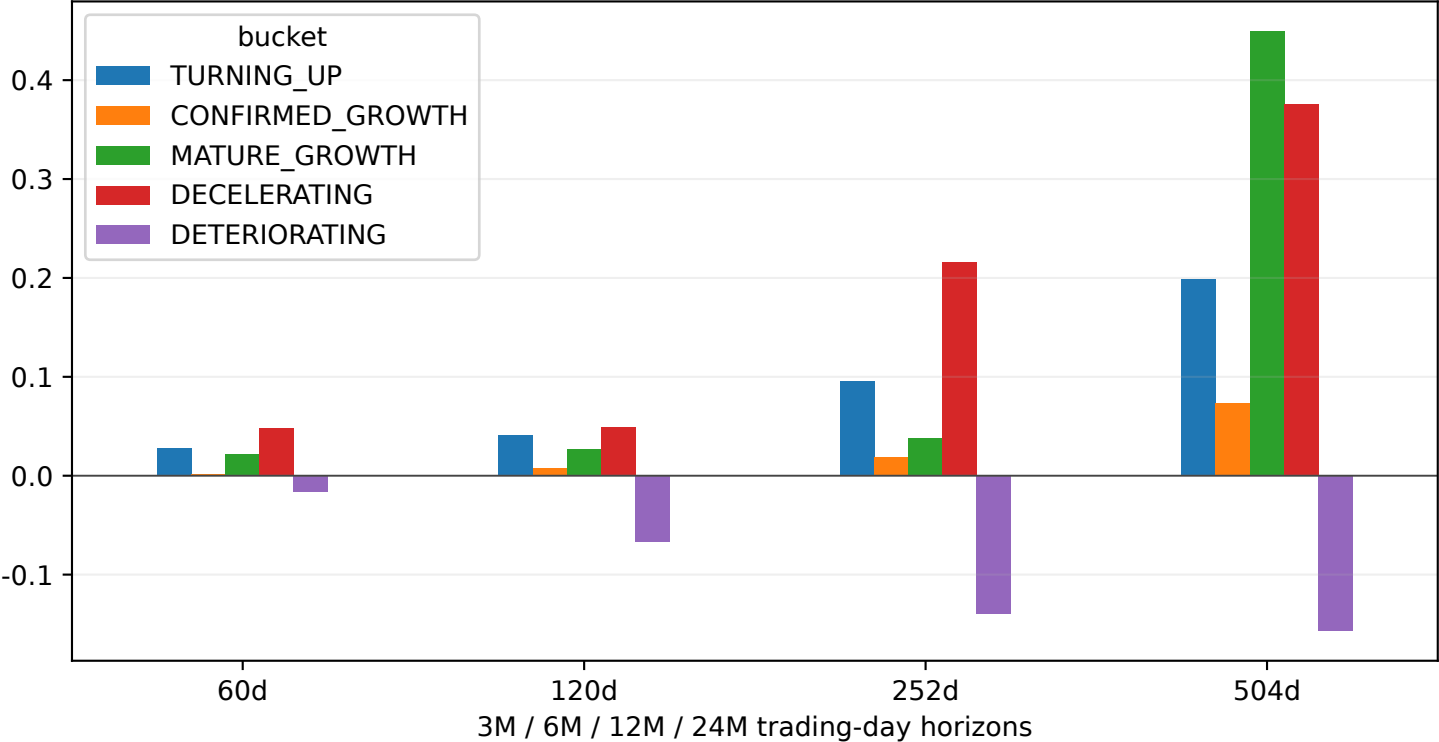
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B2. Future Return Predictive Power — state_detail

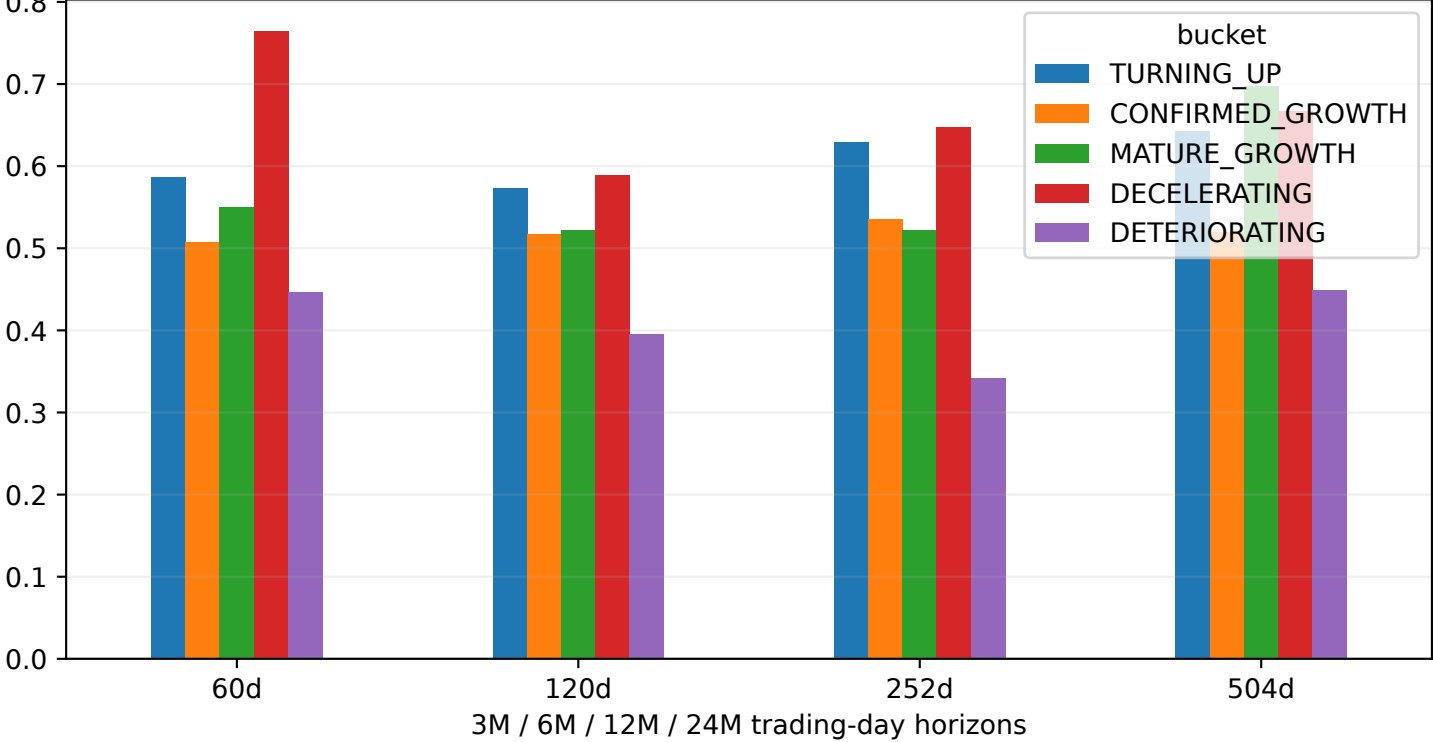
Median stock return



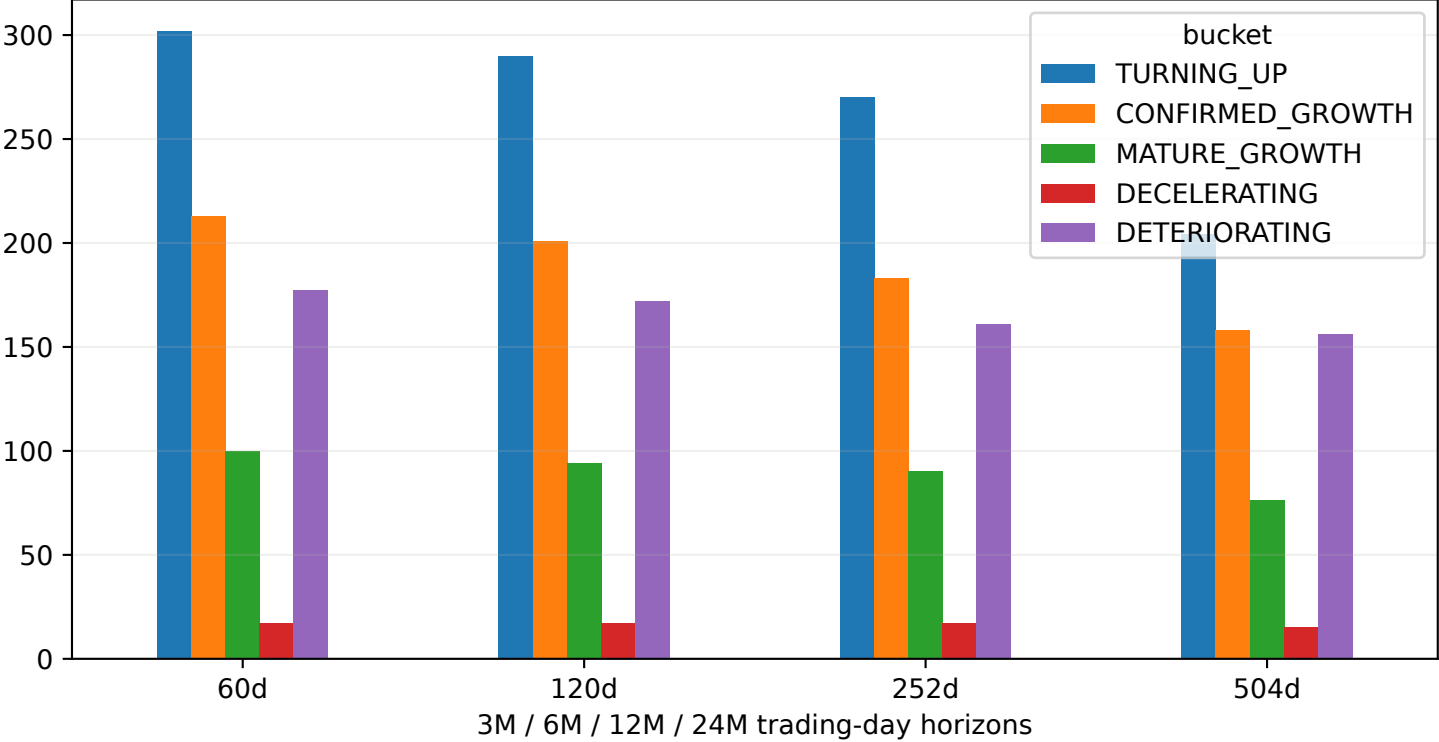
Median excess return vs 0050



Outperform-0050 rate



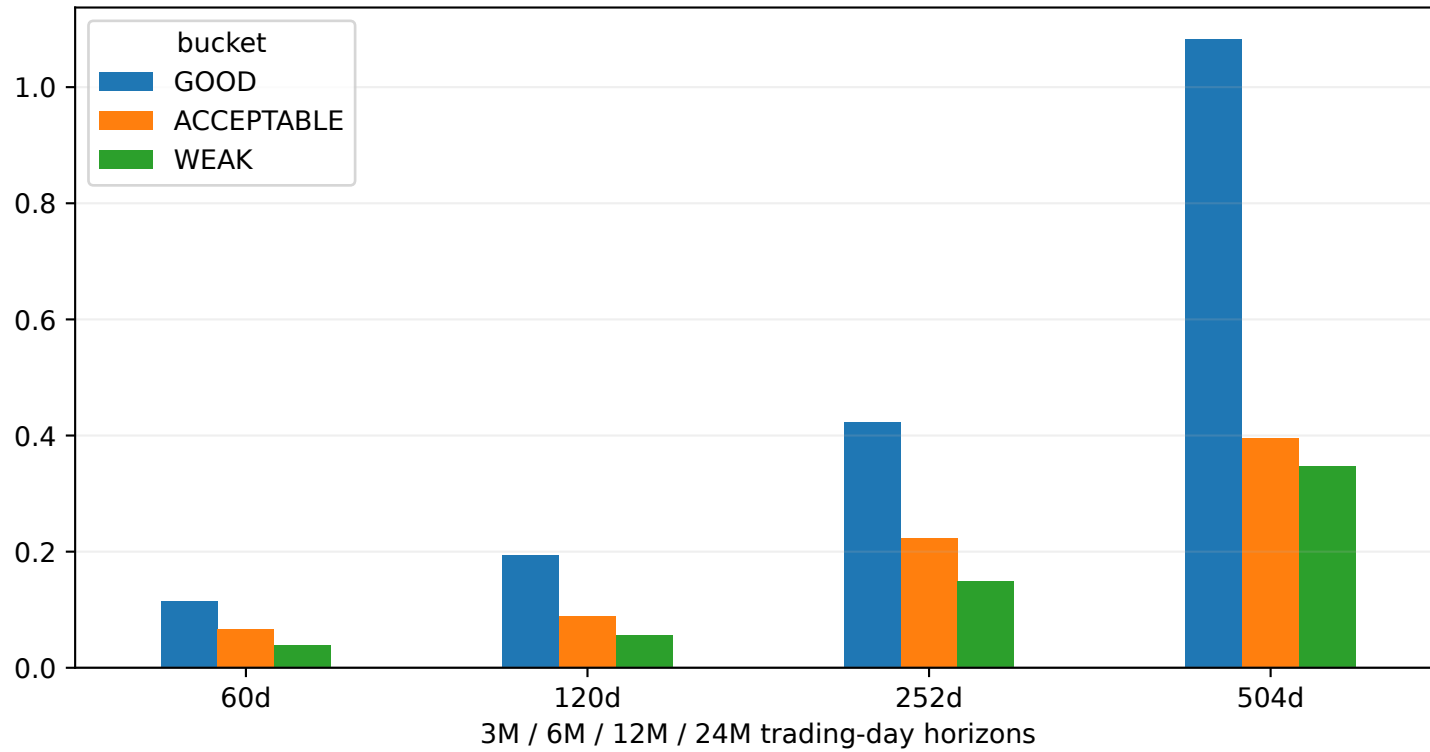
Evaluable observations



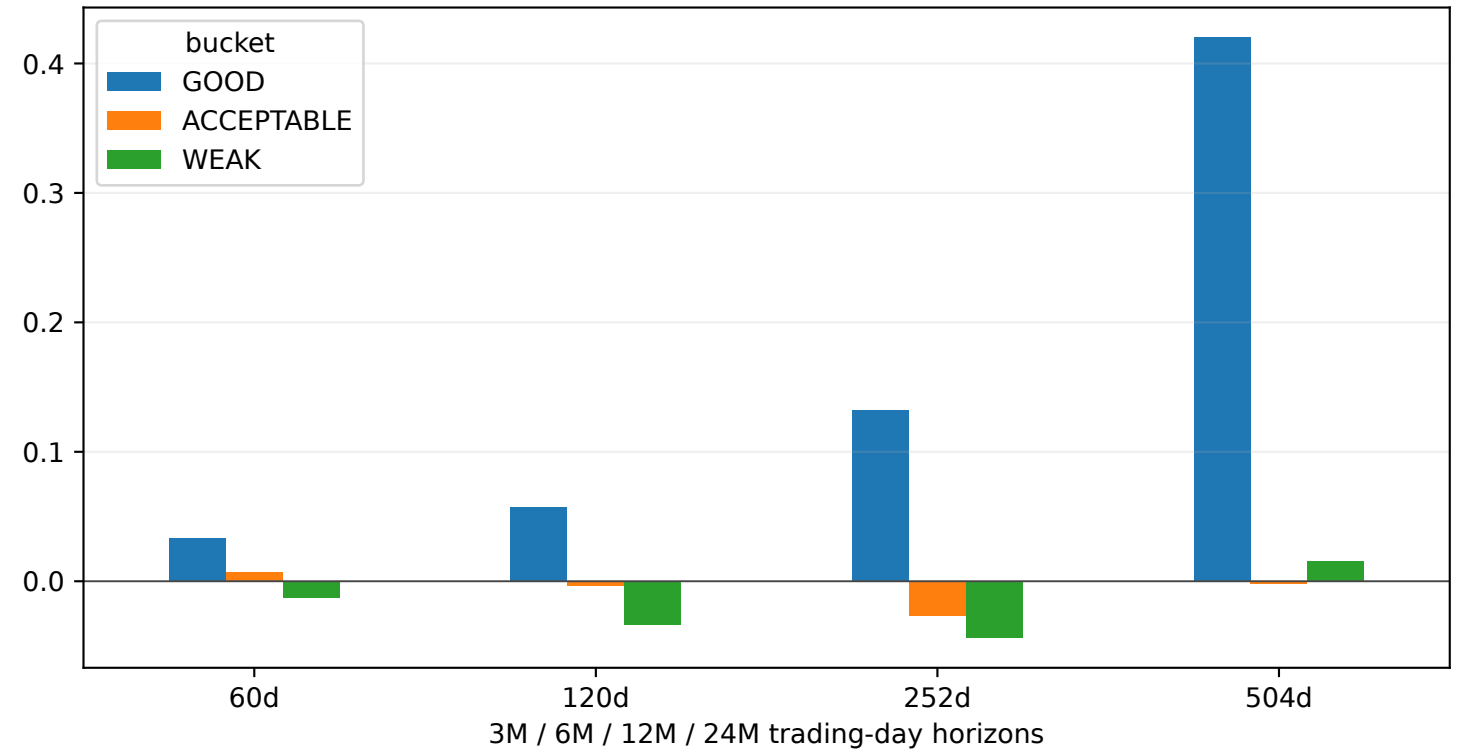
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B3. Quality Predictive Test

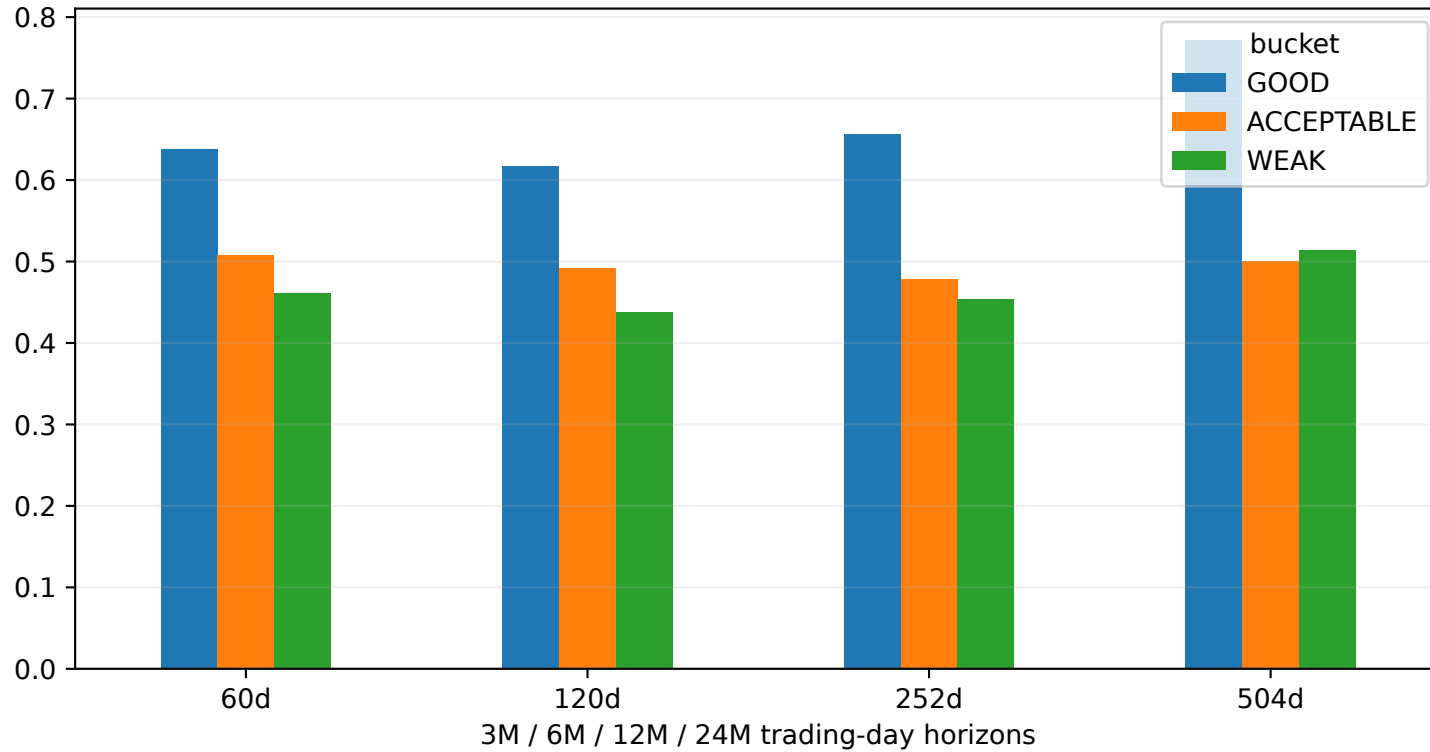
Median stock return



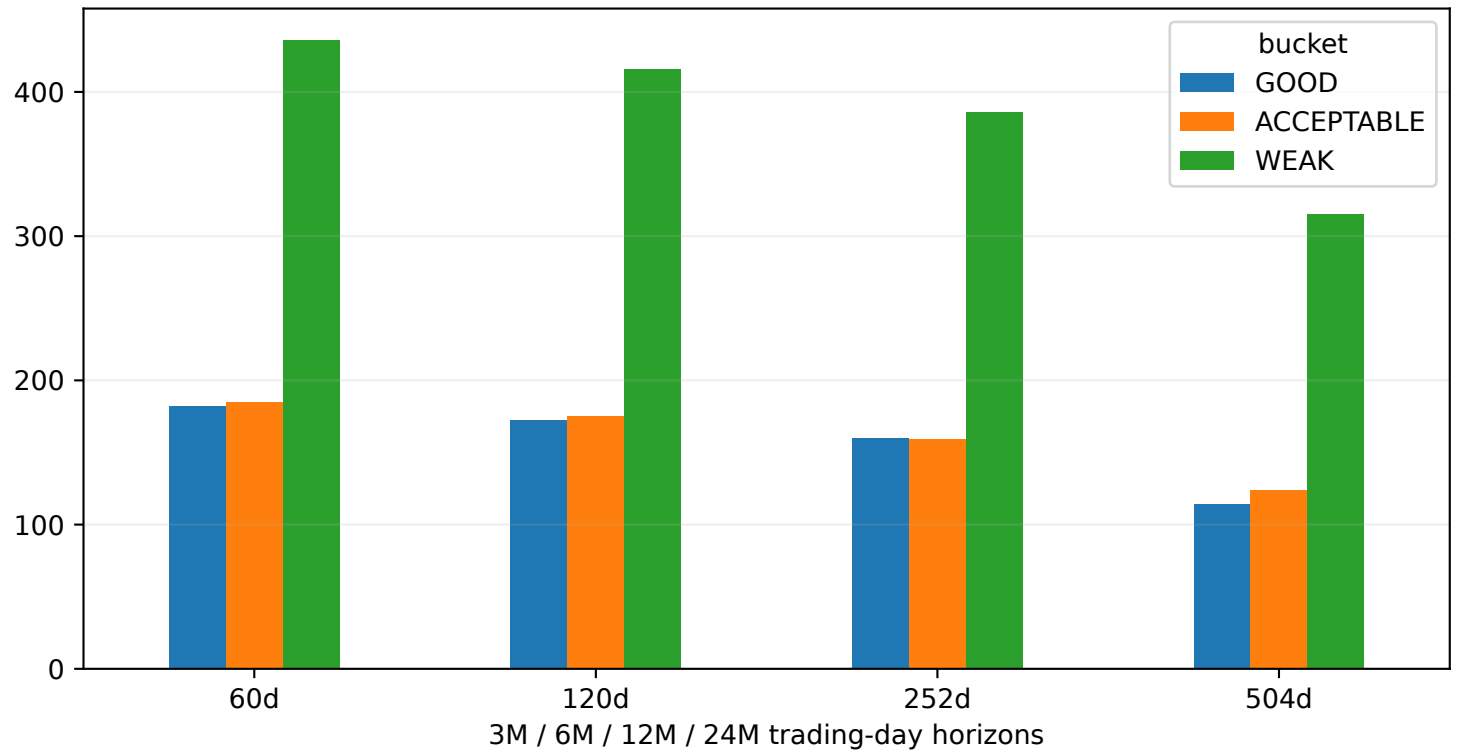
Median excess return vs 0050



Outperform-0050 rate



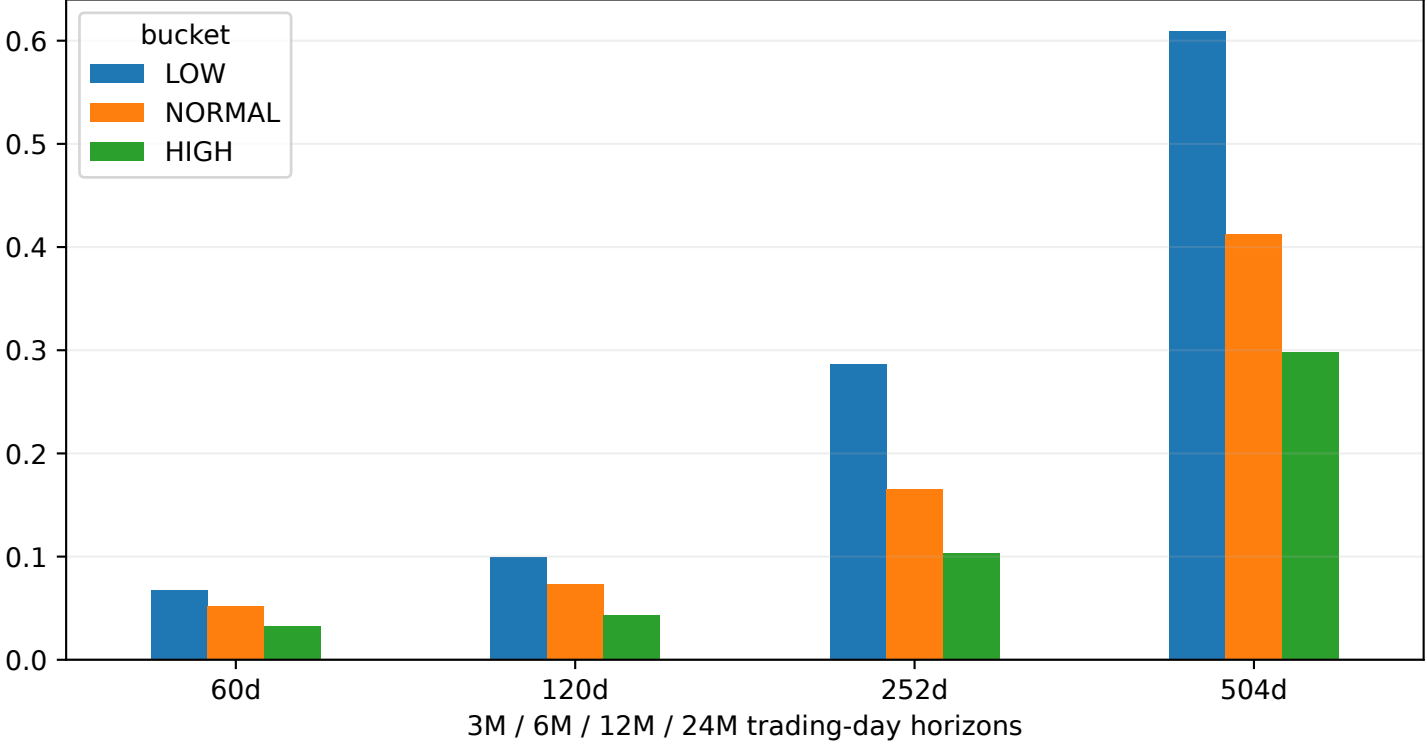
Evaluable observations



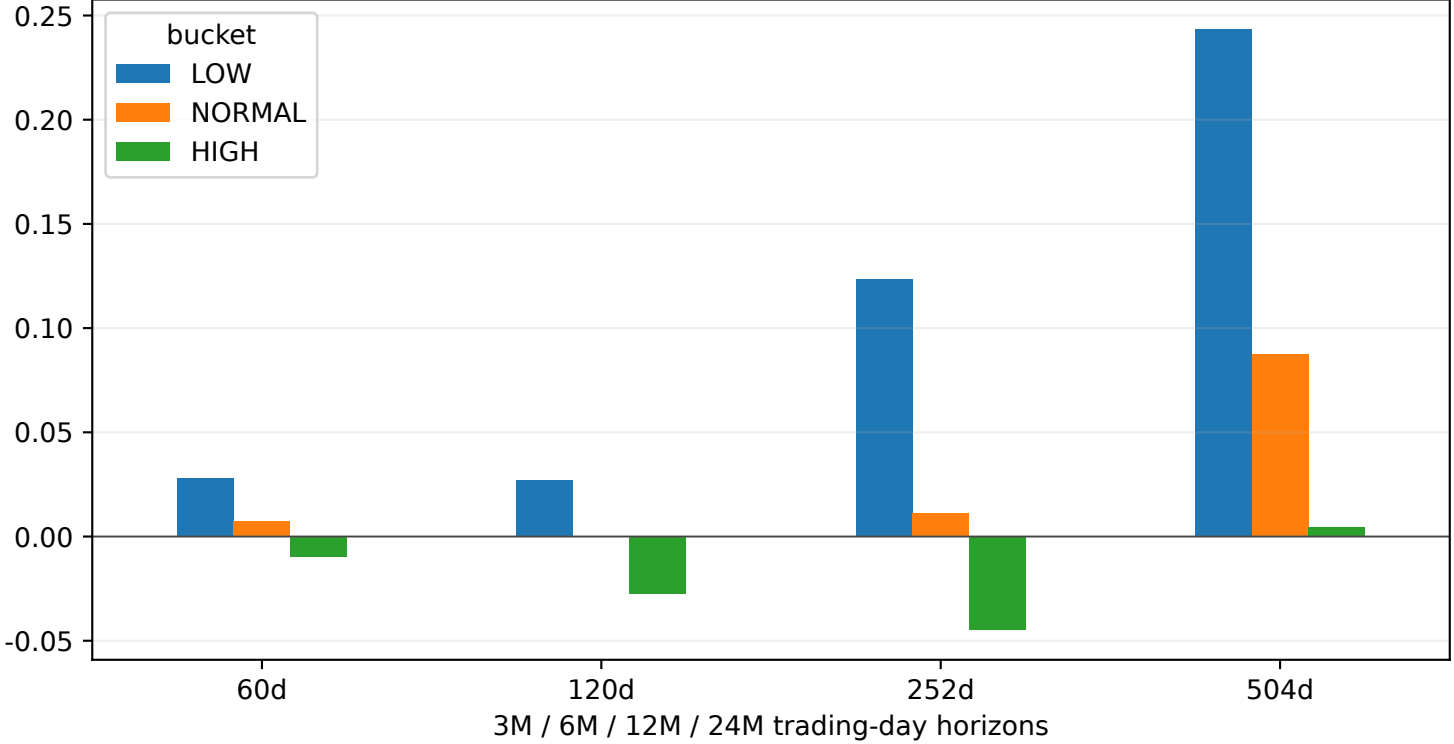
Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B4. Valuation Predictive Test

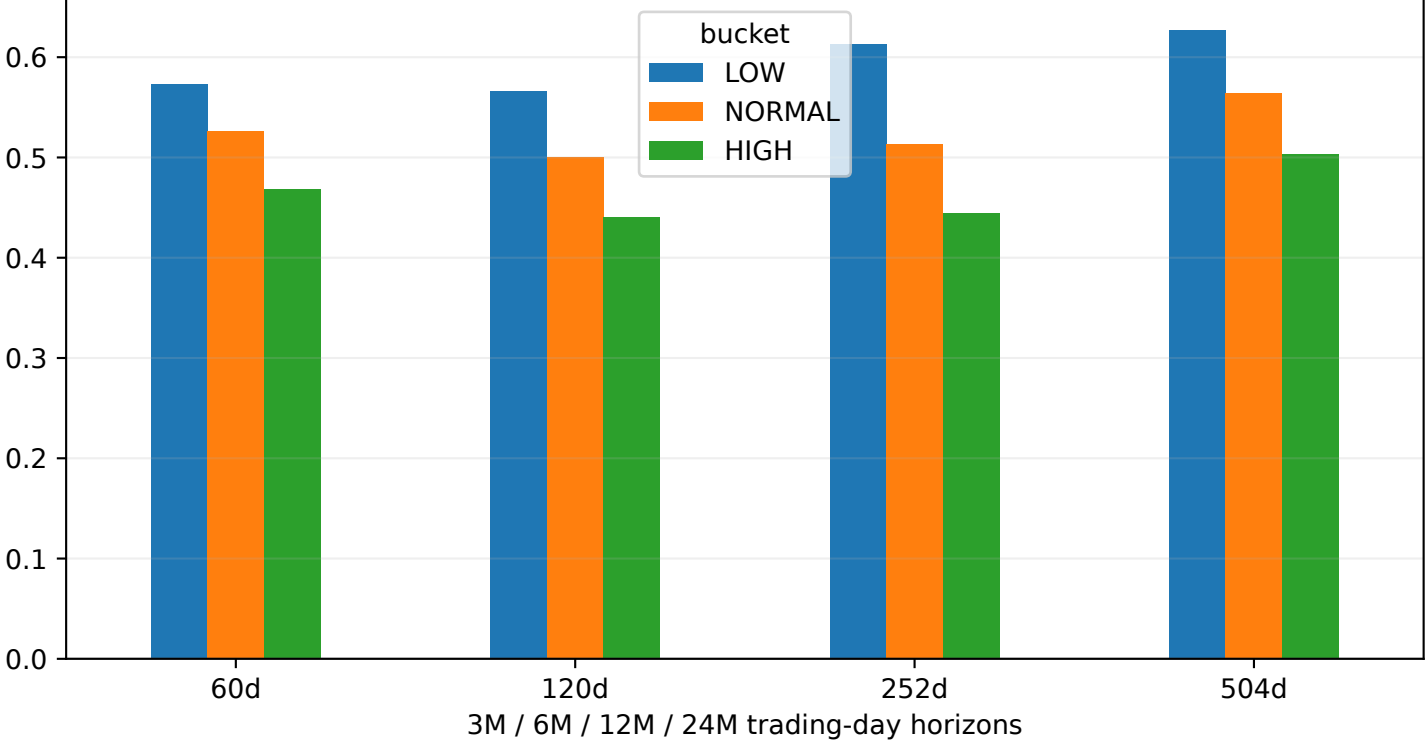
Median stock return



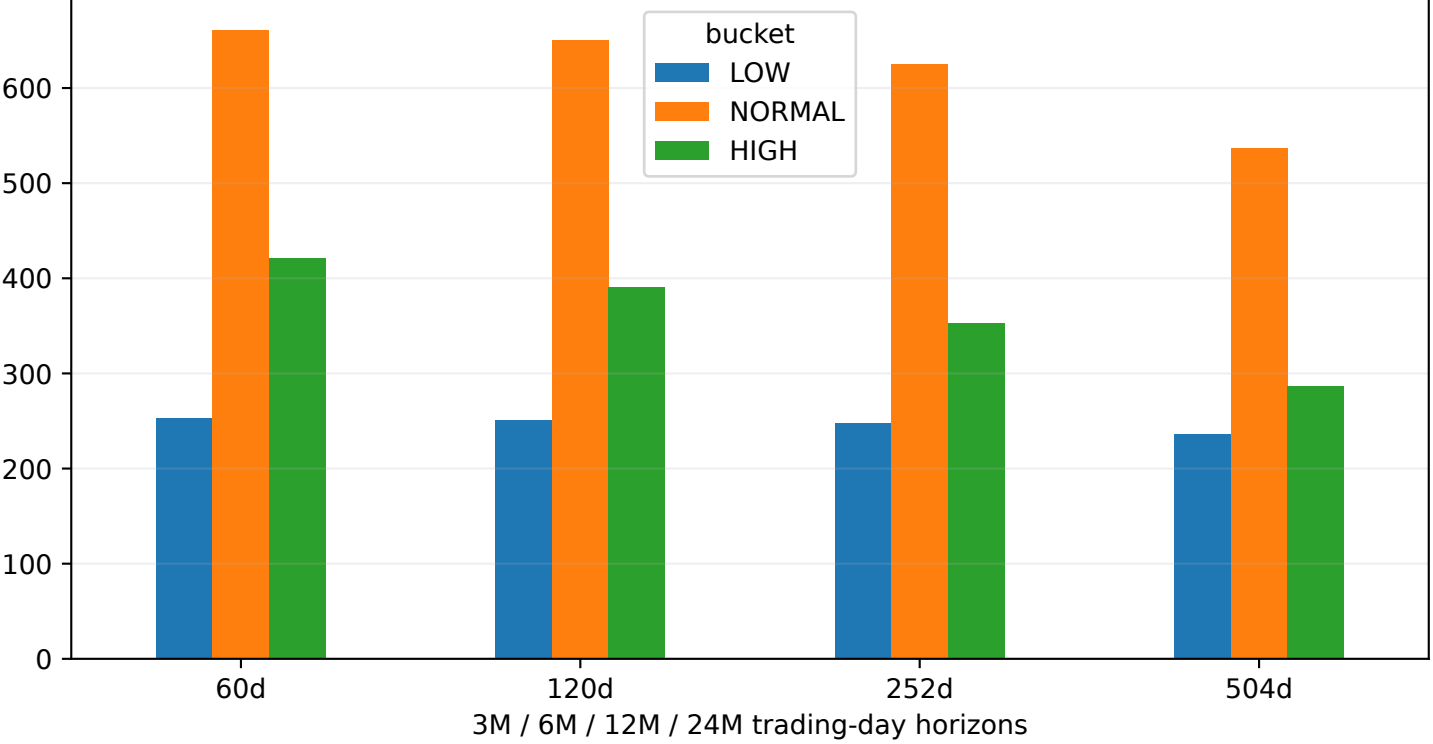
Median excess return vs 0050



Outperform-0050 rate



Evaluable observations



Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B5. Supported Categorical Intersections — 12M

Dimension	Cell	N	Issuers	Median	Median excess	Outperform	Excess lift
QUALITY_X_STATE	GOOD × STABLE	52	15	69.9%	40.1%	71.2%	38.8%
QUALITY_X_STATE	GOOD × IMPROVING	60	14	62.2%	17.3%	80.0%	16.0%
STATE_X_VALUATION	STABLE × LOW	87	24	34.9%	12.4%	62.1%	11.1%
QUALITY_X_VALUATION	ACCEPTABLE × NORMAL	78	23	30.4%	11.3%	57.7%	9.9%
QUALITY_X_VALUATION	WEAK × LOW	79	22	37.3%	10.3%	60.8%	8.9%
STATE_X_VALUATION	IMPROVING × NORMAL	140	33	28.2%	8.6%	62.1%	7.2%
QUALITY_X_VALUATION	GOOD × HIGH	64	14	28.2%	7.8%	64.1%	6.4%
STATE_X_VALUATION	IMPROVING × LOW	61	29	30.8%	7.1%	60.7%	5.7%
STATE_X_VALUATION	IMPROVING × HIGH	102	29	18.6%	5.5%	54.9%	4.2%
QUALITY_X_VALUATION	GOOD × NORMAL	70	19	38.6%	5.4%	60.0%	4.0%
STATE_X_VALUATION	STABLE × NORMAL	143	32	21.7%	2.5%	53.8%	1.2%
QUALITY_X_STATE	ACCEPTABLE × IMPROVING	49	20	22.2%	0.1%	51.0%	-1.3%
QUALITY_X_STATE	WEAK × IMPROVING	109	28	15.4%	-0.3%	49.5%	-1.6%
STATE_X_VALUATION	STABLE × HIGH	48	22	6.3%	-1.2%	47.9%	-2.6%
QUALITY_X_STATE_X_VALUA	WEAK × IMPROVING × NOR	50	22	14.1%	-2.0%	46.0%	-3.4%
QUALITY_X_STATE	ACCEPTABLE × STABLE	40	17	23.5%	-2.7%	47.5%	-4.1%
QUALITY_X_VALUATION	ACCEPTABLE × HIGH	39	11	18.4%	-2.8%	41.0%	-4.1%
QUALITY_X_STATE	WEAK × STABLE	117	25	9.3%	-4.1%	42.7%	-5.4%
QUALITY_X_VALUATION	WEAK × NORMAL	194	27	14.6%	-5.1%	44.8%	-6.4%
QUALITY_X_STATE_X_VALUA	WEAK × STABLE × NORMAL	67	21	11.7%	-9.1%	43.3%	-10.4%
QUALITY_X_STATE_X_VALUA	WEAK × IMPROVING × HIGH	31	13	6.8%	-9.9%	38.7%	-11.2%
QUALITY_X_VALUATION	WEAK × HIGH	94	19	6.7%	-11.4%	36.2%	-12.8%
STATE_X_VALUATION	DETERIORATING × NORMAL	77	29	15.2%	-12.9%	31.2%	-14.2%
QUALITY_X_VALUATION	ACCEPTABLE × LOW	42	10	21.1%	-13.3%	35.7%	-14.6%

B5a. GOOD × STABLE Disclosure

Selection status

GOOD × STABLE is the ex-post strongest supported two-axis cell at 252d among 23 supported cells.

Multiplicity

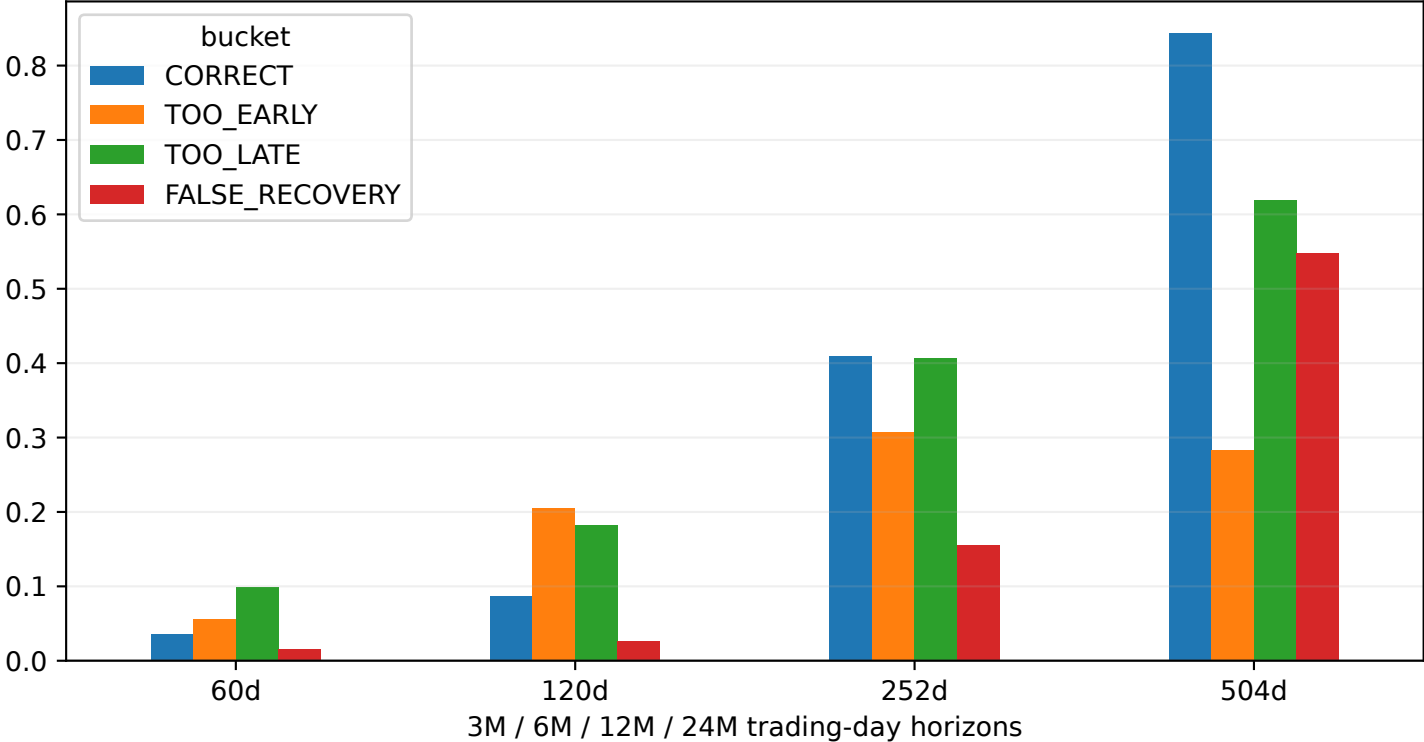
No multiple-comparison adjustment was applied. Winner's-curse risk exists.

Claim boundary

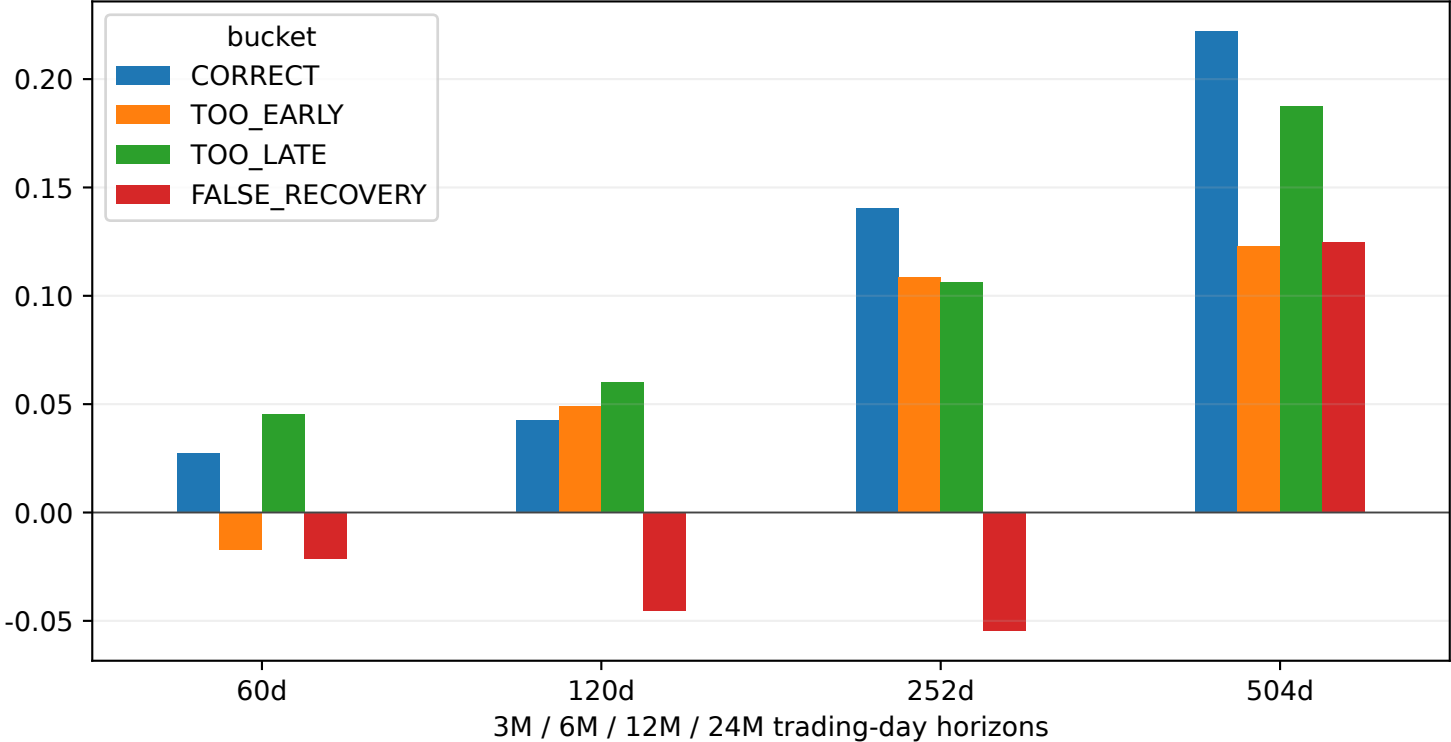
Exploratory descriptive evidence only. It is not a validated investment rule, not a pre-registered winner, and does not increase the evidence grade.

B6. TOO_LATE Investment Relevance

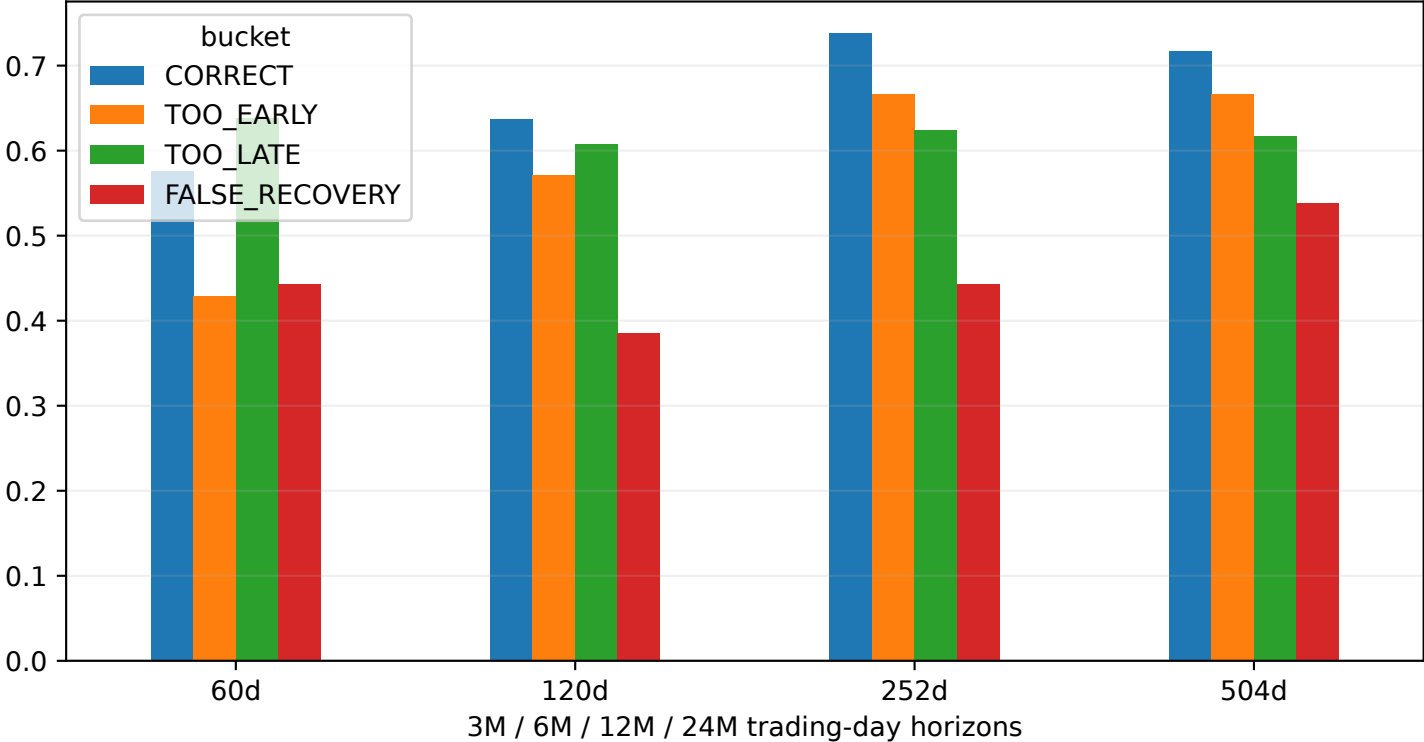
Median stock return



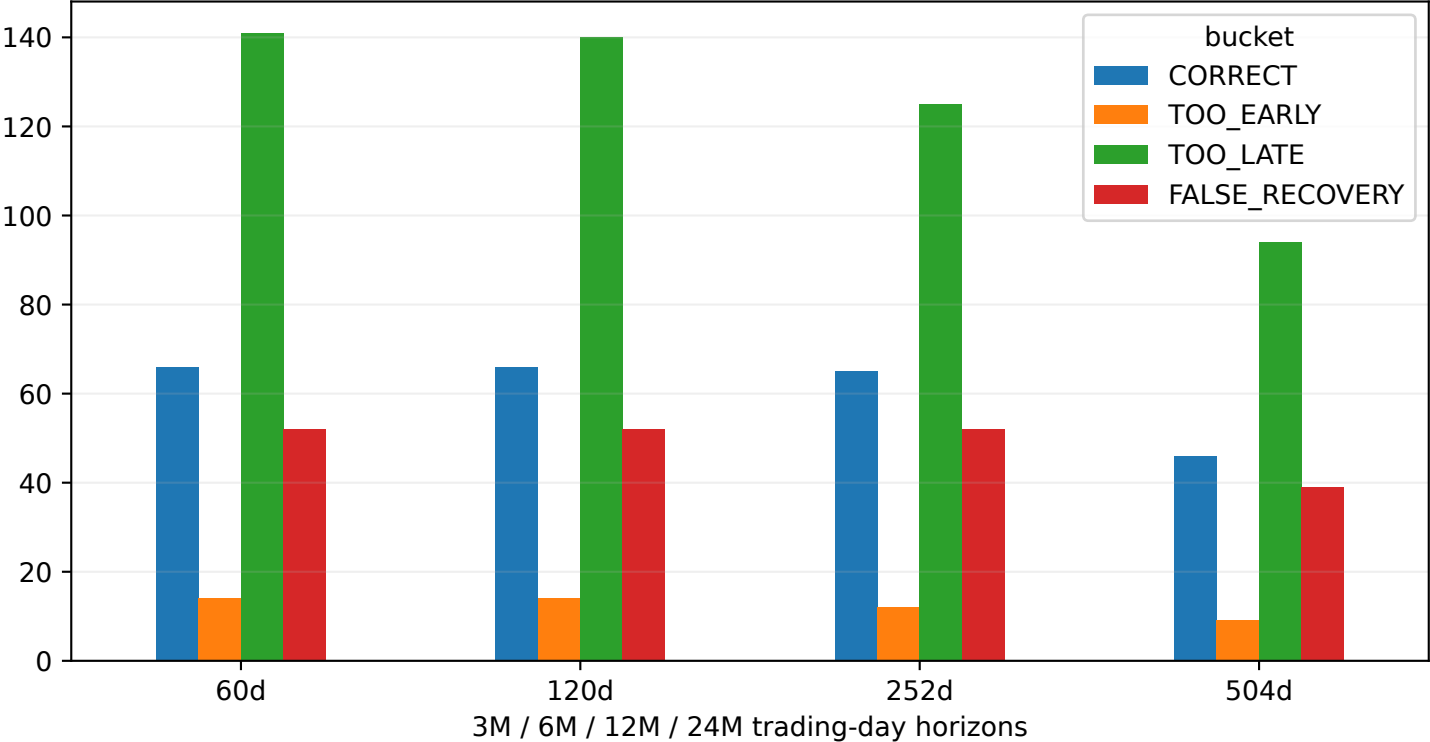
Median excess return vs 0050



Outperform-0050 rate



Evaluable observations



Means, medians, dispersion, issuer-clustered and entry-quarter-clustered uncertainty are retained in the CSV artifacts.

B7. Regime Robustness — pre-2023 vs 2023 onward

Regime	State	Horizon	N	Issuers	Median excess	Outperform	Support
PRE_2023	IMPROVING	60d	199	37	1.7%	55.3%	SUPPORTED
PRE_2023	IMPROVING	120d	199	37	2.6%	55.3%	SUPPORTED
PRE_2023	IMPROVING	252d	199	37	7.1%	62.3%	SUPPORTED
PRE_2023	IMPROVING	504d	199	37	17.6%	63.8%	SUPPORTED
PRE_2023	STABLE	60d	180	35	2.5%	58.9%	SUPPORTED
PRE_2023	STABLE	120d	180	35	4.8%	60.0%	SUPPORTED
PRE_2023	STABLE	252d	180	35	5.5%	60.0%	SUPPORTED
PRE_2023	STABLE	504d	180	35	16.5%	61.1%	SUPPORTED
PRE_2023	DETERIORATING	60d	75	31	-2.4%	42.7%	SUPPORTED
PRE_2023	DETERIORATING	120d	75	31	-6.6%	36.0%	SUPPORTED
PRE_2023	DETERIORATING	252d	75	31	-12.1%	32.0%	SUPPORTED
PRE_2023	DETERIORATING	504d	75	31	-12.4%	41.3%	SUPPORTED
2023_ONWARD	IMPROVING	60d	145	36	2.8%	57.9%	SUPPORTED
2023_ONWARD	IMPROVING	120d	134	36	4.2%	56.7%	SUPPORTED
2023_ONWARD	IMPROVING	252d	113	36	6.6%	55.8%	SUPPORTED
2023_ONWARD	IMPROVING	504d	35	23	28.7%	60.0%	SUPPORTED
2023_ONWARD	STABLE	60d	150	32	-2.6%	46.7%	SUPPORTED
2023_ONWARD	STABLE	120d	132	31	-6.3%	41.7%	SUPPORTED
2023_ONWARD	STABLE	252d	110	31	-12.1%	43.6%	SUPPORTED
2023_ONWARD	STABLE	504d	69	25	0.1%	50.7%	SUPPORTED
2023_ONWARD	DETERIORATING	60d	102	35	-1.0%	46.1%	SUPPORTED
2023_ONWARD	DETERIORATING	120d	97	35	-6.8%	42.3%	SUPPORTED
2023_ONWARD	DETERIORATING	252d	86	33	-19.6%	36.0%	SUPPORTED
2023_ONWARD	DETERIORATING	504d	81	32	-18.2%	48.1%	SUPPORTED

B8. Outlier and Concentration Robustness

Scope	Horizon	N	Issuers	Median	Median excess	Outperform	Top-5 contribution
FULL_SAMPLE	60d	344	37	6.7%	2.3%	56.4%	5.7%
EX_TSMC	60d	330	36	6.8%	2.5%	56.1%	5.7%
TOP_1_RETURN_REMOVED	60d	344	37	6.7%	2.3%	56.4%	5.7%
TOP_5_RETURNS_REMOVED	60d	343	37	6.7%	2.2%	56.3%	5.7%
FULL_SAMPLE	120d	333	37	12.3%	3.0%	55.9%	6.4%
EX_TSMC	120d	319	36	12.4%	3.8%	55.5%	6.4%
TOP_1_RETURN_REMOVED	120d	333	37	12.3%	3.0%	55.9%	6.4%
TOP_5_RETURNS_REMOVED	120d	333	37	12.3%	3.0%	55.9%	6.4%
FULL_SAMPLE	252d	312	37	27.3%	7.0%	59.9%	6.1%
EX_TSMC	252d	298	36	26.9%	7.1%	58.7%	6.1%
TOP_1_RETURN_REMOVED	252d	312	37	27.3%	7.0%	59.9%	6.1%
TOP_5_RETURNS_REMOVED	252d	312	37	27.3%	7.0%	59.9%	6.1%
FULL_SAMPLE	504d	234	37	63.8%	19.9%	63.2%	5.1%
EX_TSMC	504d	224	36	63.0%	17.9%	62.9%	5.1%
TOP_1_RETURN_REMOVED	504d	234	37	63.8%	19.9%	63.2%	5.1%
TOP_5_RETURNS_REMOVED	504d	231	37	63.0%	18.1%	62.8%	5.1%

B9. Overlapping-Window Audit

Horizon	Eligible	Overlapping obs	Ratio	Pairs	Same issuer	Cross issuer/calendar
60d	1507	1507	100.0%	49282	446	48836
120d	1462	1462	100.0%	101978	2057	99921
252d	1395	1395	100.0%	212160	5187	206973
504d	1224	1224	100.0%	333214	8461	324753

B10. Evidence Grade and Limitations

MOPS PIT coverage

Frozen Stage A financial timeline: 2,067 observations; 2,053 MOPS exact (99.3%) and 14 explicit available-date proxies. Stage B performs zero MOPS network requests and does not remap any filing date.

Evidence assessments

Mechanical rubric grade: STRONG. Independent reviewer evidence assessment: MODERATE. Primary research evidence assessment: MODERATE.

Pre-registration

NOT_VERIFIABLE. No immutable pre-result Git checkpoint independently proves the rubric was fixed before predictive outcomes were observed.

Uncertainty

Clustered inference: LIMITED. Issuer-clustered, entry-quarter-clustered, and cluster-bootstrap estimates are retained, but they are not a complete two-way dependence model.

Overlap

Overlap handling: LIMITED. Primary-horizon overlap is 100%, and long-horizon common market shocks are not fully controlled.

Targeted tests

Original: 38; correction: 15; current total: 53.

Limitations

CURRENT_CONSTITUENTS_ONLY; results are not survivorship-bias-free historical 0050 performance. Repeated issuer observations and overlapping windows are not IID. Issuer- and entry-quarter-clustered uncertainty does not eliminate all common-regime dependence. Adjusted-close paths are close-to-close; excursions are not intraday MFE/MAE. Categorical intersections are exploratory and are not scores or tuned trading rules. Financial issuers are excluded from all predictive evidence. Primary-horizon overlap is 100%; clustered inference and overlap handling remain LIMITED. Factor and sector confounding remain; marginal associations are not independent effects. Pre-registration cannot be independently verified from an immutable pre-result Git checkpoint. Multiple comparisons are unadjusted; the ex-post strongest supported cell has winner's-curse risk.

Prohibited claims

No buy/sell recommendation, target price, expected-return promise, historical-0050 strategy performance, causal claim, composite score, or parameter optimization.